Kevin T. Zyskowski

Applied ML Researcher
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Education

Master of Science in Artificial Intelligence | Northeastern University | GPA: 3.91/4.00 Sep 2023 – May 2025
- Thesis: From Input to Identity: An AI-Driven Approach to Player Identification in Counter-Strike 2

Bachelor of Science in Computer Science | Northeastern University | GPA: 3.96/4.00 Sep 2019 – Sep 2023
- Summa cum laude; concentration in Machine Learning

Projects

PlayID: Open-Set Watchlist Identification from Player Input in Deadlock Oct 2026 – Present
- Framed ban evasion as open-set watchlist identification (ISO/IEC 19795): account-disjoint splits, time-ordered enrollment, and held-out accounts as unknown probes; targeting arXiv and IJCB 2027.
- Decoded 82K public match replays (412K accounts) into per-tick camera angles and held keys; trained an ECAPA-TDNN encoder (ArcFace loss, modality dropout) with a cohort-normalized watchlist search.
- On validation (~2K-account watchlist), 2.5% FNIR at 1% FPIR vs. 61.9% for a hand-crafted baseline; 98.9% rank-1.

M.S. Thesis: Player Identification from Gameplay Telemetry Sep 2024 – Apr 2025
- Built a Playwright + Polars pipeline that scraped and parsed 1,000+ professional Counter-Strike 2 match replays from HLTV into mouse and keystroke telemetry, yielding 103K recoil-spray samples from 1,411 players.
- Trained a Siamese LSTM in PyTorch with online triplet mining over packed variable-length sequences; embeddings reached 7x chance-level 5-NN player identification (22% vs. 3%) on 31 held-out players.

Bass Transcription from Separated Audio | github.com/ktzyskowski/bass-transcription Jul 2026 – Aug 2026
- Transcribes isolated bass audio into notes; raised held-out note F1 on Demucs-separated audio from 0.74 to 0.88 (synthetic Slakh2100) through single-variable ablations, with no larger model or extra data.
- Decomposed errors into over-segmentation, voicing, and pitch buckets to locate each bottleneck; showed pYIN capped recall, then replaced it with a learned pitch head (+5.0 F1 on test).
- Built a differentiable semi-Markov CRF decoder with truncated BPTT, traced a NaN gradient from a finite forward pass, and retired it after it trailed greedy decoding by 4.1 F1 on held-out test.

DreamerV3 from Scratch | github.com/ktzyskowski/dreamer Mar 2026 – Apr 2026
- Reimplemented DreamerV3 from the paper in PyTorch, an agent that learns a model of its environment and trains its policy on imagined rollouts; solved CartPole-v1 (mean return 491 vs. the 475 threshold).
- Built the RSSM world model (categorical latents), imagination-trained actor-critic, twohot reward/value heads, and EMA critic, plus a custom environment to test the world model's memory.

Experience

AI Engineer | The Mutual Group | Remote Aug 2026 – Oct 2026
- Designed and built Imagination Hub, a full-stack idea-submission app with LLM chat (Anthropic API) on AWS.

AI Engineer | Staples Inc. | Framingham, MA (Hybrid) Sep 2024 – Jan 2026
- Fine-tuned the production retriever for Staples' product-matching platform over a 9M+ SKU catalog (MPNet, custom PyTorch loop, MNR loss, FAISS-mined semi-hard negatives), reaching 87.8% Success@10 and 75.8% Coverage@10.
- Built a 433K-pair retriever training set by merging cross-reference and competitor match data, with sampled negatives.
- Built a retrieval evaluation framework (Success@K, Recall@K, Coverage@K, MRR) for three matching use cases.
- Co-built with a colleague the embedding training pipeline on Azure Databricks, with pluggable train/test sets, MLflow tracking, and Azure Pipelines CI/CD.

Machine Learning Intern | Staples Inc. | Framingham, MA (Hybrid) Jun 2024 – Sep 2024
- Refactored the production ML inference pipeline (91.5% less code), enabling online model swapping and shipping a full test suite and docs; earned a full-time return offer.

Skills

Languages: Python, SQL
Methods: Dense retrieval, rank fusion, and re-ranking, Metric and contrastive learning, Sequence modeling (LSTM, TCN), Structured prediction (semi-CRF), Model-based RL, Audio signal processing
ML & Modeling: PyTorch, Hugging Face sentence-transformers, scikit-learn, FAISS, Gymnasium
Data: NumPy, pandas, Polars, Spark, Apache Arrow, Snowflake
MLOps & Cloud: Docker, MLflow, Azure Databricks, Azure Pipelines, AWS (Lambda, S3, EC2)